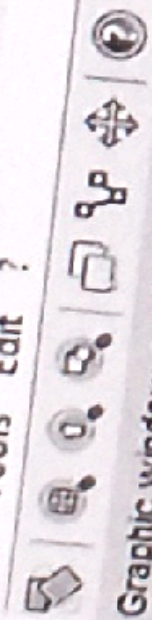
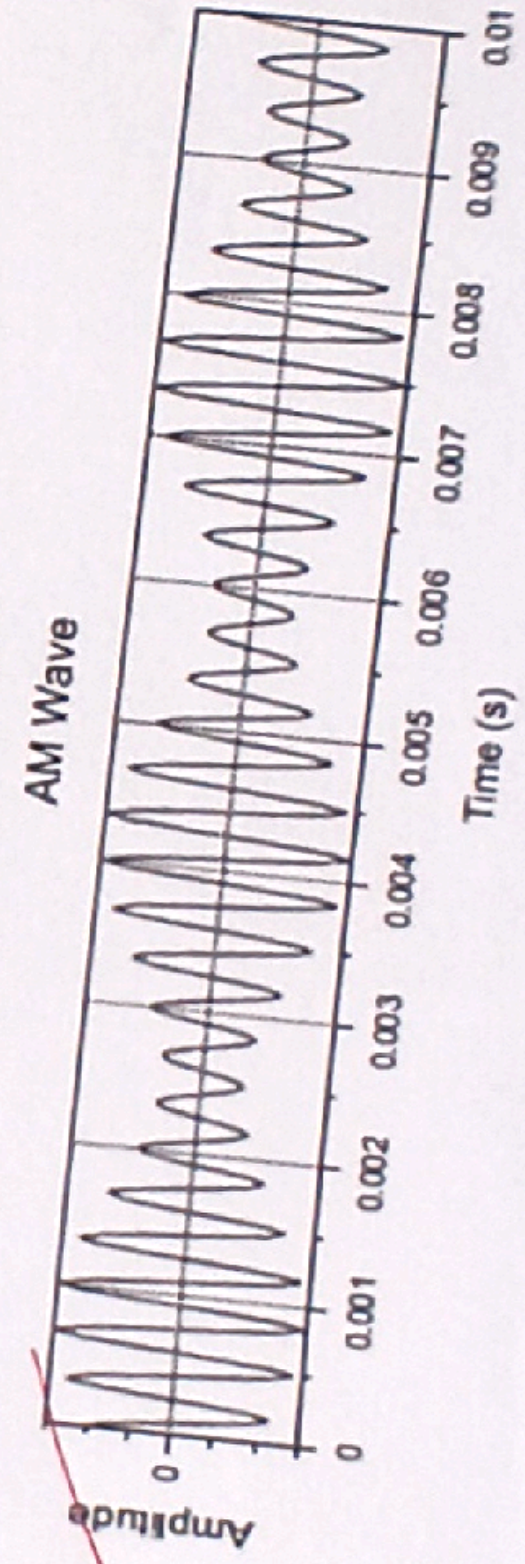
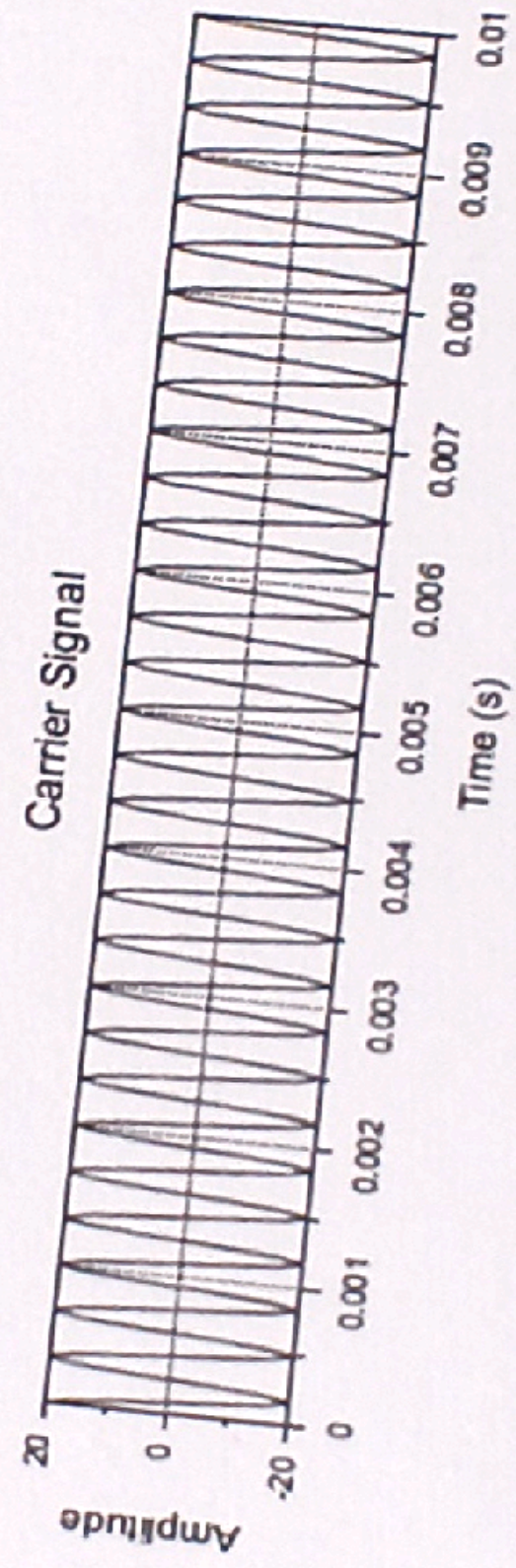
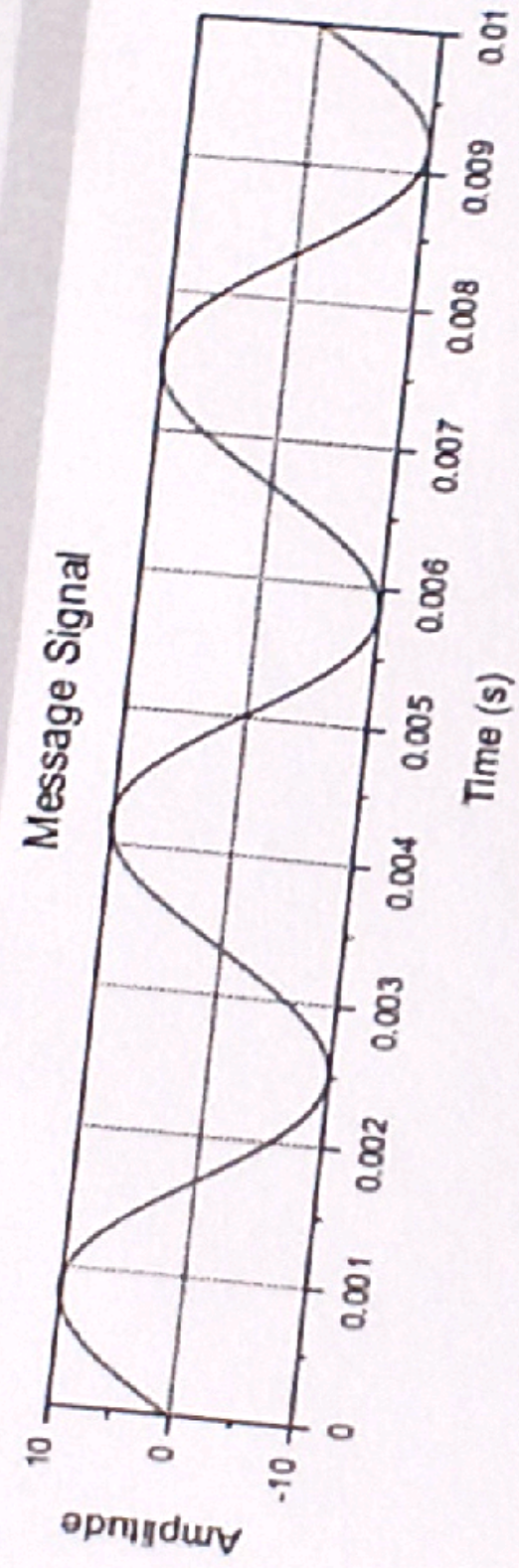


File Tools Edit ?



Graphic window number 0



20/05/26

Exp - 2 AMPLITUDE MODULATION

AIM:

To Simulate amplitude modulation (AM) and modulation index and Bandwidth analysis.

APPARATUS REQUIRED:

Scilab Software

CODE :

$E_m = 10;$

$F_m = 300;$

$E_c = 20;$

$F_c = 3000;$

$F_s = 30000;$

$t = 0:1 / F_s = 3 / F_m;$

$am = E_m * \sin(2 * \%pi * F_m * t);$

Subplot (3,1,1);

plot (t, am);

title ("Message Signal");

$c = E_c * \sin(2 * \%pi * F_c * t);$

Subplot (3,1,2);

plot (t, c);

title ("Carrier Signal");

$cam = (E_c + am) * \cos(2 * \%pi * F_c * t);$

Subplot (3,1,3);

plot (t, cam);

title ("Am Waves");

S.No	Description	Theory			Practical		
		Amplitude	Frequency	Amplitude	Time	Frequency	
1.	Message Signal	10	300	10	$T_m = \frac{1}{300}$ $= 0.00334$ $= 0.00417 -$ 0.00083 $= 0.00334s$	$f_m = \frac{1}{T_m}$ $= \frac{1}{0.00334}$	
2.	Carrier signal	20	3000	20	$T_c = \frac{1}{3000}$ $= 0.000333$ $\Rightarrow 0.00133 -$ 0.00100 $\Rightarrow 0.000333$	$F_m = \frac{1}{T_m}$ $\Rightarrow 3000$	
3.	Am Modulator Signal	-	-	$E_{max} = 30$ $E_{min} = 10$	$T_{AM} = 0.000333$ $T = x_2 - x_1$ $= 0.00183 -$ 0.00100 $= 0.000833$	$f_m = \frac{1}{T_m}$ $\Rightarrow 3000$	

CALCULATIONS:-

Theoretical ;

$$M_a = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}} \Rightarrow \frac{1}{2} = 0.5$$

$$\therefore (M_a)_{\text{Theory}} = (M_a)_{\text{practical}}$$

Scilab Program & Implementation (20)	Waveforms and Tabulation (20)	Calculation and result (20)	viva voce (20)	Result/observation (20)
20	20	20	17	20 97

RESULT:

AM waves are successfully generated thus the modulated index is calculated.

Q84